

(TR-104) PYTHON WITH AI/ML

Training Day 48 Report

Date: 18 March 2026

Topic Covered: Revision of RAG Concepts & Introduction to Streamlit

The forty-eighth day of training focused on revising important concepts related to RAG systems, fine-tuning, and guardrails, along with learning the fundamentals of Streamlit for building interactive Python web applications.

- Revised the core concepts of Retrieval-Augmented Generation (RAG) and understood the overall workflow of retrieval and response generation.
- Reviewed fine-tuning concepts and understood how trained models are adapted for task-specific applications.
- Revised the purpose and implementation of guardrails for improving AI system safety and preventing unsafe responses.
- Started learning Streamlit for developing interactive and user-friendly Python web applications.
- Practiced displaying text elements using `st.title()`, `st.header()`, `st.subheader()`, `st.text()`, and `st.markdown()`.
- Explored different methods of displaying structured data using `st.write()`, `st.table()`, and `st.dataframe()`.
- Added media elements such as images using `st.image()`.
- Worked with interactive widgets including checkboxes, radio buttons, buttons, select boxes, and multiselect options.
- Implemented file upload functionality using `st.file_uploader` for both single and multiple files.
- Explored sliders using `st.slider` for numerical input handling.
- Collected user input using `st.text_input()`, `st.text_area()`, `st.date_input()`, and `st.time_input()`.
- Learned how widget events work using `on_click` and `on_change` functions.
- Explored session management using `st.session_state` for maintaining application state.
- Understood the difference between `text_input` and `text_area` in Streamlit applications.

- Strengthened understanding of how Streamlit simplifies frontend development for AI and machine learning applications.
- Improved conceptual understanding of integrating user interaction with AI-powered systems.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/streamlit>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/my-first-streamlit-app-interactive-web-application-using-python-2f8fc6933870>